

Muhammad Shoaib (PMP)®

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SUMMARY

I am an experienced computer professional proficient in Python, machine learning, statistical analysis, desktop application development, and AI Agent development, capable of delivering reliable data-driven solutions. I possess excellent problem-solving skills and a strong mathematical foundation, with expertise in leveraging Large Language Models (LLMs) to build automated workflows and intelligent information extraction systems. I am highly adaptable, speak multiple languages, and consistently strive for excellence. In addition, I hold a Project Management Professional (PMP) certification, demonstrating my ability to effectively lead and coordinate projects.

TECHNICAL SKILLS

Languages	Python, R, SQL, HTML/CSS, JavaScript
ML / AI	Machine Learning (Classification, Regression), Deep Learning, PyTorch, Transfer Learning, LLM-based AI Agents
AI Agents	AI Agent Development, LLM Automation, LinkedIn Profile Extraction, Workflow Orchestration
Neuro / Bio	Neuroimaging (fMRI/MRI), CIFTI/NIFTI, Nilearn, nibabel, CIBERSORTx, Seurat, snRNA-seq, AHBA
GUI / Web	PyQt5, Tkinter (Desktop Apps), Django (Web Development)
Data	Pandas, NumPy, Matplotlib, Seaborn, MySQL
Robotics	YOLOv7, SLAM, ROS, RK3588 Platform, PID Control, STM32
Tools	Git, HCP Workbench (wb_command), LibreOffice, VS Code

EDUCATION

Northwestern Polytechnical University | M.Sc. **Computer Science & Technology** Sep 2024 – Mar 2027

Master's Thesis Research:

- Developed the Single Cell Annotation Toolbox (SCAT) a Python/PyQt5 desktop application integrating AHBA-derived cell type spatial maps with disease-specific functional connectivity (FC) networks meta-analysed across eleven psychiatric and neurological disorders. The toolbox provides three analytical modules: a 3D FC Network Viewer with cell type overlays on Schaefer parcellations (100–1000 parcels), a Cell Type Enrichment Analysis module using spatial permutation testing (up to 10,000 permutations) to identify significantly enriched cell types in disorder-affected brain clusters, and a 3D Results Viewer projecting z-score enrichments onto interactive brain surfaces with HTML export. Applied to Parkinson's disease data, SCAT identified endothelial cell enrichment in the lingual gyrus and co-localization of GABAergic interneuron density with dysregulated FC hubs.

Northwestern Polytechnical University | B.Sc. **Computer Science & Technology** Sep 2020 – Jul 2024

Outstanding Graduate Award (2024)

- Bachelor's Thesis: Applied advanced machine learning techniques including Support Vector Machines (SVM) and Random Forest classifiers to analyze fMRI and MRI data, improving diagnostic accuracy and efficiency for schizophrenia. Employed rigorous 10-fold cross-validation to ensure robust discrimination between schizophrenia patients and healthy controls.

WORK EXPERIENCE

Shanghai Yugou Technology (上海宇构科技) | AI Engineer Intern

Jun 2025 – Sep 2025

- **Pet AI Platform:** Trained photo-based pet emotion recognition, dog & cat breed classification, and cat behaviour recognition models using transfer learning. Developed audio-based cat vocalization behaviour recognition and dog vocalization mood recognition models. Integrated all models into a full-stack pet website.
- **Food Recognition:** Trained EfficientNet-B0 on diverse food images for real-time recognition, deployed the model on-chip for camera-based live food identification, and generated nutritional summaries (protein, carbohydrates, fat).
- **Robotics:** Deployed YOLOv7 on an RK3588 platform for real-time small-pet detection and target tracking. Integrated LiDAR with SLAM-based mapping for autonomous indoor obstacle avoidance; used STM32 with PID control for smooth motor-driven target following.

AI Agent — LinkedIn Profile Extractor | Online Project

- Built an LLM-powered AI Agent that automatically extracts LinkedIn profiles matching user-defined criteria (skills, department, and geographic location). The agent integrates automated web scraping, natural language instruction parsing, and structured data output, enabling efficient and precise talent search and information aggregation for recruitment and research workflows.

Concordia College | Web Developer

Aug 2021 – Dec 2021

- Designed and developed a school grade management system by integrating front-end and back-end solutions to meet institutional requirements. Created a user-friendly platform that streamlined academic workflows, improved grade management efficiency, and enhanced data accessibility.

Sage Education Ltd. | IT Support Intern

Jun 2022 – Aug 2022

- Provided hands-on troubleshooting, diagnostics, and resolution of technical issues to ensure operational efficiency.
- Managed network operations, optimizing connectivity and system performance across the organization.
- Implemented cybersecurity measures, monitored protocols, and enhanced system protection.
- Led successful upgrades of internal software systems, improving functionality and user experience.

CERTIFICATIONS, AWARDS & LANGUAGES

 Project Management Professional (PMP)®

 English: Duolingo 130/160

 NPU Outstanding Graduate Award (2024)

 Chinese: Intermediate HSK 3

 NPU Future AI Masters Int'l Summer Camp Certificate

 Skillvention Python Programming Certification

 University of Michigan: Programming for Everybody (Python)

 University of Michigan: Data Structures Certificate
